Scale now

This is the session-start brief. Read this page first. Read the roadmap archive or the decision records only by section, on demand; they are history, not a briefing.

State

- Example suite: **1530 checks** (1488 examples, 42 properties), zero failures at the last verification.
- Granular suite: 33 cases (11 snapshots, 13 integrations, 9 writer checks); discovery is 91 tests.
- ABI and image schema: version 2.
- Polylith: 11 components, 2 bases, 3 projects, 0 development, 1 external.
- Last verified checkpoint: Slices 1–6, R3, R4, R6, R8, R10, R11, R12, and R13 are landed. The canonical renames and verification move are complete; the shell policy enforces allowed programs, write roots, and environment choice; the semantic test matrix is recorded.
- Architecture and the specification status matrix are generated from `docs/architecture.edn`; `just docs-check` runs the generator in check mode.
- Current work: none. The remaining security work is explicitly future: read paths, network policy, and package signatures.
- Next: release-time gates only (`just check`, `just aot-test`, `just release`).

Read order

1. This page.
2. `AGENTS.md` for the workflow and the tool policy.
3. `docs/specification.typ` by section for the feature under change.
4. The feature's `.typ` (for example `docs/prose-as-shell.typ`).
5. `docs/irs.typ` when touching parsing, lowering, the image, or execution.
6. `docs/decision-index.typ` to find the active decision, then its section.
7. `docs/performance.typ` before optimizing.

`docs/roadmap.typ` is the current pointer page; `docs/roadmap-archive.typ` is completed history. Do not read the archive or `docs/decision-records.typ` end to end. Use `doc_get_structure` and `doc_get_section`, or a narrow search.

Active decisions

- D183: shell `run`/`capture` are host words; `components/shell-host` owns the process effects.
- D177 is superseded for prose: logic definitions belong in books, not in the prose language. D184 lands books/spells, `--book`, `--cast`, and the prose `cast` host word.
- D182: Scale-only Serena/F# work uses the sparse `Scale-code` workspace. The full Nu checkout is expected to fail the F# LSP project load.
- D185: canonical vocabulary. An IR is an **abstraction** a `-Ast` is a `-Logic`; `TermIr` is `SymbolicAbstraction`; `TypeInference` is `TypeRetrieval`. The docs use the canonical names; the code keeps its current names until a rename is scheduled. The `verification` directory is the canonical target for tests, formal verification, and type checking.
- D186: Basic English is the historical name for the Prose sentence grammar. The docs are `docs/prose-grammar.typ` and `docs/prose-foundations.typ`; the sentence parser remains `tokenizer.fs`/`parser.fs` and is not merged into `prose-parser.fs`.
- D190: the lowered instruction set is `LoweredInstructions`; `LoweredAbstraction` is the execution abstraction. The old module name is retired. No behavior changed.

- D191: `RootKind`/`rootKinds` is the one execution-root registry; `getRoot`/`setRoot` reattach a root. Roots remain execution state and are not serialized.
- D192: image and package schema 2 carry `Unit`/`UnitGroup`, arity, captures, initialization order, and classification; ABI 2.
- D193: group calls resolve through the unit's group, frame arguments and captures, and use a separate closure callback; short calls and missing captures are named refusals. Value-unit initialization is Slice 3.
- D194: a book is one recursive group; self, forward, and mutual calls run. `logic` and `cycle` are checked against the book calls; `and` groups cycle definitions.
- D195: core `logic` and `cycle` definitions use the plain name for recursion; `logic` refuses a self-call, and `cycle ... and ...` stores a mutual group. There is no `recurse` word.
- D196: the F# abstraction, manifest tiers, FCS read-off, supported body translation, diagnostics, metadata, and package lowering are implemented.
- D197: `ClassificationWire` carries `(name, kind, version)` only; a body, duplicate name, unknown kind, and version skew are refused.
- D198: a book package is lazy; units carry per-definition hashes and dependencies, calls check arity, `main` takes no arguments, and the book cache is keyed by resolved path plus source hash.
- D199: `open math` imports `math.book` beside the opening book; its exports are callable, its capabilities are unioned, and cycles/conflicts are refused.
- A SageFs session keeps serving the last successful build. Check `sourceState` before trusting an eval or a test; see `docs/sagefs.typ`.
- `--prose-typecheck` does not infer host words yet; run a shell script with `--prose`.